

Skill Cortex V1

Webinar / Course Booking, Payment & Notification Platform

Project objective: Users select a department, choose a webinar and available slot, pay through Razorpay, receive confirmation notifications, and receive automated reminders before their booked session.

1. Final User Workflow

Step	User Action	System Action
1	Register / Login	Authenticate user and load profile
2	Select Department	Show relevant webinars
3	Select Webinar	Display details, price and slots
4	Select Date & Time	Validate slot availability
5	Make Payment	Create Razorpay order and open checkout
6	Payment Verification	Backend verifies payment before confirming booking
7	Confirmation	Send user and admin email/SMS with amount and booking details
8	Daily Reminders	Scheduled service checks upcoming paid bookings
9	Webinar	User attends the booked session

2. System Architecture

```
USER / ADMIN
↓
REACT + TAILWIND FRONTEND
↓
FASTAPI BACKEND
■■■ Authentication
■■■ Department Management
■■■ Webinar Management
■■■ Slot Management
■■■ Booking Management
■■■ Payment Service → RAZORPAY
■■■ Notification Service → EMAIL / SMS
■■■ Reminder Scheduler
↓
POSTGRESQL DATABASE
```

3. Database / ER Structure

Core relationship: User → Department → Webinar → Slot → Booking → Payment → Notifications

Table	Important fields	Purpose / relationship
users	id, name, email, phone, password_hash, department_id	User profile; belongs to department; creates bookings
departments	id, name, description	Groups users and webinars
webinars	id, title, description, price, duration, status, department_id	Courses/webinars offered by admin
slots	id, webinar_id, date, start_time, end_time, capacity, available	Available webinar sessions
bookings	id, user_id, webinar_id, slot_id, status, created_at	Connects user to webinar slot

Table	Important fields	Purpose / relationship
payments	id, booking_id, razorpay_order_id, razorpay_payment_id, razorpay_order_id	Verified payments, recorded
notifications	id, user_id, booking_id, type, channel, message, status, timestamp	Send out confirmations and reminders

4. User Portal

- **Authentication:** registration, login, logout, password reset.
- **Profile:** name, email, phone and department.
- **Webinar catalog:** browse courses and view description, duration and price.
- **Slot booking:** choose a valid date/time with available capacity.
- **Payment:** pay securely using Razorpay Checkout.
- **My bookings:** view upcoming, completed, cancelled and payment status.
- **Notifications:** receive confirmation and reminder messages.

5. Admin Portal

- Admin authentication and protected dashboard.
- Create/update/activate/deactivate departments.
- Create webinars with title, description, price, duration and department mapping.
- Create and manage webinar dates, times, capacity and availability.
- View users and bookings.
- View payment records, amounts, Razorpay IDs and payment status.
- View notification history and delivery status.

6. Razorpay Payment Flow

User selects slot → backend creates Razorpay order → user completes checkout → Razorpay returns payment details → backend verifies payment → payment becomes PAID → booking becomes CONFIRMED → confirmation notifications are sent.

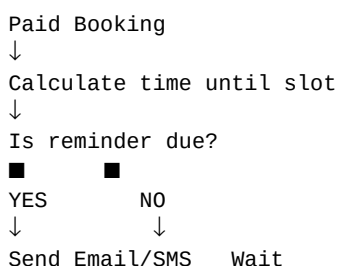
Security rule: The frontend must not be the final authority for payment success. Backend verification is required before confirming a booking.

7. Notification Flow

Event	User	Admin
Successful verified payment	Email/SMS with booking, amount, webinar and slot	Email/SMS with user, webinar, amount and payment status
Upcoming webinar	Scheduled reminder based on booked slot	Optional admin notification

8. Reminder Engine

A scheduled background job checks paid bookings whose webinar slot is approaching. It checks notification history before sending so duplicate reminders are avoided.



9. Recommended Technology Stack

Layer	Technology
Frontend	React + Tailwind CSS
Backend	Python + FastAPI
Database	PostgreSQL + SQLAlchemy
Authentication	JWT + secure password hashing
Payment	Razorpay
Email	SMTP / transactional email provider
SMS	SMS provider to be selected during implementation
Scheduler	APScheduler initially; Celery + Redis if scale requires it
Deployment	Docker + VPS
Version Control	Git + GitHub

10. Recommended Project Folder Structure

```
skill-cortex/  
  ■■■■ backend/  
  ■ ■■■■ app/  
  ■ ■ ■■■■ main.py  
  ■ ■ ■■■■ config.py  
  ■ ■ ■■■■ database.py  
  ■ ■ ■■■■ models/  
  ■ ■ ■■■■ schemas/  
  ■ ■ ■■■■ routes/  
  ■ ■ ■■■■ services/  
  ■ ■ ■■■■ utils/  
  ■ ■ ■■■■ scheduler/  
  ■ ■■■■ requirements.txt  
  ■ ■■■■ .env  
  ■■■■ frontend/  
  ■ ■■■■ src/  
  ■ ■ ■■■■ components/  
  ■ ■ ■■■■ pages/  
  ■ ■ ■■■■ services/  
  ■ ■ ■■■■ App.jsx  
  ■ ■■■■ package.json  
  ■■■■ docker-compose.yml  
  ■■■■ README.md  
  ■■■■ .gitignore
```

11. Development Phases

Phase	Deliverable
1	Architecture + database/ER design
2	FastAPI + PostgreSQL setup
3	Authentication and user roles
4	Departments, webinars and slots
5	Booking and seat availability
6	Razorpay order + payment verification
7	Email/SMS notification service
8	Automated reminder scheduler
9	Admin dashboard
10	User frontend
11	Testing, validation and security
12	Docker + VPS deployment

V1 Definition of Done

A user can register, select a department, choose a webinar and available date/time slot, complete a Razorpay payment, and receive verified confirmation. The admin can manage the catalog and view bookings/payments. The reminder service automatically notifies users according to their booked slot without duplicate reminders.